🔗 Topic 5 (Treatment Rationale) · papers 1 and 5 at a glance

- **Kokich 2004, labial canines.** *Which uncovering technique should the orthodontist ask the surgeon for?* Decide on **4 criteria**: labiolingual position, vertical position against the **MGJ**, amount of gingiva, mesiodistal position. **Excision** only if the crown is coronal to the MGJ with enough gingiva; **apically positioned flap** when gingiva is short or the crown lies mesially over the lateral root; **closed eruption** for a high or mid-alveolar crown.
- **Kokich 2004, palatal canines.** *When and how?* Extract the primary canine early if the crown is not past the **mesial surface of the lateral root**. Otherwise **uncover before brackets**, remove bone **down to the CEJ** and let the canine erupt alone (**about 6–8 months**), then align. He claims shorter treatment and better periodontal and esthetic results; later RCTs found open exposure **equivalent to closed exposure, not superior** (VERIFIED, abstracts).
- **Stephens 2005, the profile.** *Do extraction and nonextraction faces age differently when both finish at the same incisor position and lip line?* **No.** Over about **15 years (≈15 → 30 y)** both flattened the same way: lips **≈1.5 mm** further behind the E-line, profile straighter. The authors attribute it to **nose and chin growth**, much larger in males (upper lip **−3.2 vs −0.9 mm** to the E-line).
- **Stephens 2005, perception.** *Do people see it, and do they mind?* 105 orthodontists and 225 laypeople saw a **small-to-moderate change (33.9–39.2 %)**, but it did not match the ceph changes (**1 of 18** correlations) and there was **no preference pattern** (about 30 / 30 / 30 %).

Surgical and Orthodontic Management of Impacted Maxillary Canines

by **Vincent G. Kokich** · Am J Orthod Dentofacial Orthop 2004;126(3):278–283 · University of Washington

🎯 Study purpose

To review the management of impacted maxillary canines: which **uncovering technique** and **timing** the orthodontist should choose and **tell the surgeon**. If not instructed, the surgeon may choose an inappropriate technique: a difficult, sometimes lengthy eruption and an unpredictable esthetic result (p 278).

🔬 What the article is (design)

- **Narrative review (symposium paper)**, section "Advances in Orthodontics & Dentofacial Surgery". Presented at the **AAO/AAOMS Symposium, 6–8 February 2004**, Palm Springs; submitted and accepted June 2004.
- **No methods, sample or statistics.** 13 references; **5 single-patient cases** (Figs 1–5, 30 panels, derived). Level 5 (expert opinion; our grading).

📌 General facts (p 278)

- After the **third molars**, maxillary canines are the **most commonly impacted** permanent teeth (Bass 1967).
- About **⅓ labial or within the alveolus, ⅔ palatal** (Johnston 1969): 2 : 1 palatal to labial (derived).
- **Labial impaction causes:** ectopic migration of the crown **over the lateral incisor root**, or a **midline shift** leaving too little space.
- **Before surgery:** Williams (1981) suggested extracting the **primary canine as early as 8 or 9 years** (labial or intra-alveolar canine); Olive (2002): **open space** and it may erupt on its own. If both fail, uncover surgically.

🛠️ The three uncovering techniques (labial / intra-alveolar canine)

Technique	What is done	Use when	Avoid when
Excisional uncovering (Fig 1)	Tissue over the crown cut away, a simple procedure	Crown labial, **coronal to the MGJ**, enough gingiva	Crown **apical** to the MGJ: no gingiva left over the labial surface; too little gingiva (Fig 3); crown mesial over the lateral root
Apically positioned flap, APF (Vanarsdall & Corn; Figs 2, 3)	Flap moved apically: exposes the crown and **increases the gingiva width** (Fig 3B)	**Too little gingiva** (the only technique that predictably makes more); crown **mesial over the lateral root**	Crown **significantly apical** (high): unstable crown, **reintrusion** after treatment
Closed eruption (Kokich & Mathews; Fig 4)	Labial flap raised, enough bone around the crown removed for eruption without impinging on bone (Fig 4D); then erupted with a ballista loop (Fig 4E–G).	**High** or **intra-alveolar** crown	Crown mesial over the lateral root (generally not recommended)

🧭 Key "result": the four selection criteria (pp 278–279)

Criterion	Finding → technique
1 · Labiolingual position	**Labial:** any of the 3 (little if any bone over the crown). **Centre of the alveolus:** excision and APF harder (extensive labial bone removal) → closed eruption (Fig 4)
2 · Vertical position vs the MGJ	**Mostly coronal:** any of the 3 (Fig 1). **Apical:** no excision (Figs 2, 3). **Significantly apical:** no APF either → **closed eruption**, which gives adequate gingiva and no long-term reintrusion
3 · Amount of gingiva	**Insufficient:** APF is the only technique that predictably makes more gingiva (Fig 3). **Enough to leave ≥2–3 mm of attached gingiva over the crown after eruption:** any of the 3
4 · Mesiodistal position	Crown **mesial, over the lateral incisor root:** APF, complete exposure needed to move it through the alveolus (Fig 2); closed eruption or excision generally not recommended

📐 "Above / below the MGJ": reading the figure legends

- The text says **apical / coronal** to the MGJ; the legends of Figs 2–4 say **above / below**. In the maxilla, **above = apical (high)** and **below = coronal**.
- **Fig 3** crown is "partially below" the MGJ: APF was chosen for **insufficient gingiva** (criterion 3), not for height.

⚙️ Orthodontic mechanics and stability (pp 279–281)

- **Mimic normal eruption.** After closed eruption, erupt the tooth into the **centre of the alveolar ridge** (normal labial gingiva), e.g. with a **ballista loop** (Fig 4): a simple, unobtrusive **vertical** force.
- Avoid pulling it labially: risk of **bony dehiscence**, accelerated migration of the labial gingival margin and **labial recession**.

- **Crown mesial over the lateral root** (APF, Fig 2): **elastomeric chain toward the centre of the edentulous ridge**; the canine was then moved distally (Fig 2E).

📊 Labial evidence: Vermette et al 1995 (APF vs closed eruption)

Outcome	APF	Closed eruption
Gingival index, plaque index, pocket depth, bone level	**No significant difference** between the 2 techniques	(same)
Crown length	**Longer than normal**: gingival margin migrates apically	Similar to the contralateral tooth
High labial canines	**Tend to reintrude** after treatment	Not observed
Why reintrusion	The flap heals to the adjacent **mucosa**; eruption draws the mucosa coronally; afterwards it **pulls the crown apically**	Becker 2002: similar favourable esthetics (**central incisors**)

Radiographic localisation (p 281): buccal object rule (Richards) on **2 periapicals** at slightly different horizontal angles. The buccal object moves **opposite** to the beam; a palatal crown moves the **same** way: SLOB = Same Lingual, Opposite Buccal.

🦷 Palatal impaction (pp 281–283)

- **The most common impaction orthodontists see. Ericson & Kurol 1988:** extracting the **primary canine** early gives self-correction with **high predictability** if, on periapicals, the canine crown overlies the lateral incisor root but is **not past its mesial surface**. **Well beyond** the mesial surface: no self-correction; it must be uncovered and aligned.

	Conventional	Early (Kokich & Mathews)
Timing	After the first **6–9 months** of alignment, space opened first	**Before** orthodontic treatment; in some cases the **late mixed dentition**
Surgery	Crown uncovered; if too little bone is removed, the canine seems not to move (p 282)	Full-thickness mucoperiosteal flap; all bone over the crown removed down to the CEJ; flap replaced with a **hole** over the crown; dressing occasionally if high in the palate
Eruption	Dragged soon after surgery toward the space, **across the lingual of the lateral incisor roots**	**Erupts on its own** to the occlusal plane in **about 6–8 months**; then bracket, root moved labially, crown translated into the arch
Cited outcome	**Woloshyn (n = 32):** bone more apical on the **distal of the lateral** and **mesial of the canine** vs contralateral; lateral and canine root resorption typical; judges could identify the treated canine	**Schmidt (thesis):** better bone and attachment levels on canine and lateral; **little to no lateral root resorption**; hard to tell which canine was impacted

Why a palatal canine "will not move" (p 282). The crown often touches the lingual of the central and lateral incisor roots. Once the follicle is removed, the tooth **cannot resorb bone** over the crown efficiently: **enamel has no resorbing cells**, so bone goes only slowly by **pressure necrosis**. It looks ankylosed, but **ankylosis is uncommon** ("low", no figure given); the usual cause is **insufficient bone removal** (Proffit Ch 15 agrees, VERIFIED, study notes).

🧠 Main conclusions

1. The orthodontist must **choose the technique and instruct the surgeon**.
2. **Labial canines:** use the 4 criteria; excision only coronal to the MGJ with enough gingiva; APF for short gingiva or a mesial crown; closed eruption for a high or intra-alveolar crown.
3. Erupt into the **centre of the ridge** (ballista loop); **avoid labial pull**. APF on high canines → long crowns and reintrusion.
4. **Palatal canines:** extract the primary canine early when the crown is not past the mesial surface of the lateral root; otherwise **uncover early, remove bone to the CEJ, let it erupt autonomously**, then align: claimed **shorter treatment, superior periodontal and esthetic results**.

⚠️ Limitations (our reading)

- **Expert opinion** with 5 successful cases: no measurements, no n of its own, no statistics.
- The main palatal claim rests on an **unpublished thesis**. Published as **Schmidt & Kokich 2007** (VERIFIED, abstract): treated canine and adjacent lateral roots **shorter** than controls, lower crestal bone at the lateral, treated canine identified **78.9 %** of the time; better for the canine than closed exposure, **similar for adjacent teeth**.
- **Timing and surgery change together** (younger patients, different era), so age is confounded with technique; the **treatment-time** claim cites no data.
- **Vermette:** 30 maxillary **anterior** teeth, not canines only (APF 18, closed 12; VERIFIED). **Becker** studied central incisors.
- **Reporting note (p 283):** judges could identify the treated canine because its tissue "compared **favorably**": the two halves point opposite ways; "unfavorably" fits Woloshyn's ~75 % identification (VERIFIED, abstract).

🔄 What later evidence and Proffit say

Kokich's claim	Later evidence (VERIFIED, abstracts)	Verdict
Primary canine extraction "will" normalise it	Naoumova 2015 RCT **69 % vs 39 %**; Bazargani 2014 **67 % vs 42 %** (better if younger; lost arch perimeter → palatal arch); Ericson & Kurol: 91 % distal vs **64 %** mesial to the lateral-root **midline** (VERIFIED, secondary)	**Helps**, not guaranteed
Open early exposure superior	Parkin 2013 RCT attachment **MD 0.1 mm** (−0.2 to 0.5); Cochrane 2017 success **RR 0.99** (0.93–1.06)	**Equivalent**
Treatment time reduced	Björksved 2021 RCT total time **MD −0.1 months** (−3.2 to 2.9)	**No difference**

🩺 Clinical implications

Scenario	Ask the surgeon for
Labial, coronal to MGJ, gingiva OK	Any; excision simplest (Fig 1)
Labial, too little gingiva	**APF** (Fig 3)
Labial, mesial over the lateral root	**APF** + chain to ridge centre (Fig 2)
High or mid-alveolar crown	**Closed eruption** + ballista loop (Fig 4)
Palatal, not past the lateral's mesial surface	**Extract the primary canine**
Palatal, beyond it	**Early exposure** to the CEJ (Fig 5)

✅ Number	Meaning
⅓ : ⅔	labial or intra-alveolar : palatal
8 or 9 y	Williams: extract primary canine
3 / 4	uncovering techniques / selection criteria
≥ 2–3 mm	attached gingiva after eruption: any technique
2	periapicals, different angles (SLOB)
6–9 months	alignment before conventional uncovering
6–8 months	canine uncovered to the CEJ reaches occlusal plane
32	Woloshyn patients (dragged canines)

Long-Term Profile Changes in Extraction and Nonextraction Patients

by **Stephens, Boley, Behrents, Alexander and Buschang** · Am J Orthod Dentofacial Orthop 2005;128(4):450–457 · Baylor College of Dentistry and Saint Louis University

🎯 Study purpose

To find whether **extraction and nonextraction patients treated to the same incisor position and lip line** show different **long-term soft-tissue profile changes**, and whether **orthodontists and laypeople perceive** and **prefer** those changes.

Background (pp 450–451): a debate of almost 100 years: extraction "dishes in" the face vs nonextraction "blows out" the lips. At debond extraction faces are on average **2–4 mm flatter** (refs 3–7); a change that is measured but not perceived is questionable.

🔬 Study design

- **Retrospective matched cohort + perception survey. 40 white, Class I and Class II** patients, **one orthodontist**, one private practice: **20 four-premolar extraction, 20 nonextraction**, each **6 M + 14 F**.
- **Matched at posttreatment** on age, sex and **3 ceph values**: **U1-SN** and **L1-NB** (dental protrusion, chosen from a discriminant analysis) and **Holdaway's H-line** (lip protrusion independent of nose length). The extraction group was picked first with values "in the normal range"; nonextraction patients were matched to it.
- **Ceph:** Dentofacial Planner, **15 landmarks**, x–y coordinates from **posttreatment sella** on a reference line **S-N −7°**; h = parallel, v = perpendicular; **9 measurements** (Table II). 15 patients replicated: no systematic error; method error 0.3–1.4.
- **Survey:** 2 versions of **20 patients** each, posttreatment vs long-term profile photos side by side (left/right random, 4 pairs per page). Raters: **105 orthodontists, 115 laywomen, 110 laymen** (330, derived); mailed surveys 60 % reply. **106-mm VAS** "same" → "very different" (distance ÷ 106 = % change perceived), plus a preference: posttreatment, long-term or neither.
- **Stats:** t tests (groups), paired t (changes), single-sample t (perception), Pearson r, chi-square.

Table I (years, mean ± SD)	Ext M (6)	Nonext M (6)	Ext F (14)	Nonext F (14)
Posttreatment age	15.6 ± 1.5	15.1 ± 1.5	14.8 ± 1.1	15.1 ± 2.2
Long-term age	30.3 ± 3.4	30.1 ± 5.3	28.9 ± 5.0	30.0 ± 4.4
Interval	14.7 ± 2.5	15.0 ± 4.9	14.1 ± 4.6	14.9 ± 2.9

Measurements and method error (Table II): N'-Pn-Pg' total facial convexity 1.40° · N'-Sn-Pg' facial convexity 1.07° · Ls-E 0.34 · Li-E 0.29 (upper / lower lip to Ricketts' E-line, mm) · Ls-S 0.44 · Li-S 0.29 (Steiner's S-line) · **H-line = perpendicular distance from Holdaway's H-line to the superior labial sulcus** 0.28 mm · U1-SN 1.45° · L1-NB 0.44 mm.

📊 Key results 1: extraction vs nonextraction (Table III, means)

Variable	Posttreatment Ext / Non (P)	Long-term Ext / Non (P)	Change Ext / Non (P)
N'-Pn-Pg' (°)	131.2 / 132.2 (.33)	132.1 / 132.1 (.99)	0.9 / −0.2 (.29)
N'-Sn-Pg' (°)	163.6 / 163.5 (.95)	167.6 / 166.4 (.47)	**+4.0* / +2.9*** (.33)
LS-E (mm)	−5.9 / −5.4 (.37)	−7.3 / −7.0 (.63)	**−1.5* / −1.6*** (.82)
LI-E (mm)	−3.6 / −4.2 (.42)	−4.9 / −5.7 (.36)	−1.3* / −1.6* (.66)
LS-S (mm)	−2.2 / −2.4 (.73)	−3.2 / −3.5 (.49)	−1.0* / −1.2* (.70)
LI-S (mm)	−1.2 / −2.2 (.15)	−2.1 / −3.4 (.10)	−0.9* / −1.2* (.56)
H-line (mm)	3.7 / 3.6 (.88)	3.7 / 3.2 (.19)	0.1 / −0.4 (.21)
U1-SN (°)	105.6 / 103.8 (.38)	105.1 / 104.9 (.93)	−0.6 / 1.1 (.23)
L1-NB (mm)	3.9 / 3.4 (.42)	3.9 / 3.1 (.24)	0.1 / −0.3 (.32)

* significant change within the group (P < .05). N'-Pn-Pg' is printed "N'-Pr-Pg'" in Tables III–IV. Negative lip values = lips **behind** the line; a negative change = **more retrusive**.

- **No group difference** at posttreatment, at long term, in the change, or in any landmark's horizontal or vertical movement (text, p 453). **5 of 9** measures changed in **both** groups: the 4 lip lines and N'-Sn-Pg'. H-line, N'-Pn-Pg', U1-SN and L1-NB did not.

📐 Facial convexity: the angle went up, so the profile got straighter

- N'-Sn-Pg' is the angle at subnasale; **180° = a straight profile**. It rose **+4.0° / +2.9°** (and **+5.5° in males vs +2.6° in females**): the face became less convex, flatter, as the Discussion says (convexity "decreased considerably", p 455; orthodontists noticed change "as the profile straightened", p 456) and as the abstract says for males ("flatter").
- The Results text calls the same change an "increase in facial convexity": read it as an **increase in the angle**. Say "the angle increased, the profile straightened".

👫 Key results 2: males vs females (Tables IV–V)

Variable (12 M / 28 F)	Posttreatment M / F (P)	Long-term M / F (P)	Change M / F (P)
N'-Sn-Pg' (°)	164.4 / 163.2 (.48)	169.8 / 165.8 (.02)	**+5.5* / +2.6*** (.02)
LS-E (mm)	−5.8 / −5.5 (.69)	−9.0 / −6.4 (< .01)	**−3.2* / −0.9*** (< .01)
LI-E (mm)	−4.7 / −3.5 (.13)	−7.6 / −4.3 (< .01)	**−3.0* / −0.8*** (< .01)
LS-S / LI-S (mm)	−2.4 / −2.3 (.85); −2.4 / −1.4 (.19)	−4.3 / −2.9 (.02); −4.4 / −2.1 (< .01)	−1.9* / −0.7* (.03); −2.0* / −0.7* (.01)
H-line (mm)	4.0 / 3.5 (.23)	3.8 / 3.3 (.16)	−0.1 / −0.2 (.87)
U1-SN (°)	107.0 / 103.8 (.15)	109.6 / 103.0 (.01)	+2.6* / −0.8 (.02)
L1-NB (mm)	3.0 / 3.9 (.17)	2.3 / 4.0 (.02)	−0.7 / 0.1 (.08)

N'-Pn-Pg' did not differ by sex (change 0.6 / 0.3°, P = .79). Sexes were alike at posttreatment.

Table V, change (mm)	Forward (h): M / F (P)	Downward (v): M / F (P)
Pn (nose tip)	**6.7* / 3.1*** (.01)	2.7* / 0.4 (.01)
Columella / subnasale	4.9* / 2.6* (.05); 3.5* / 1.5* (NS)	3.0* / 0.8 (.03); 2.3 / 0.7 (NS)

Table V, change (mm)	Forward (h): M / F (P)	Downward (v): M / F (P)
Ls (upper lip)	**4.1* / 1.8*** (.05)	3.5* / 1.5* (.04)
Li (lower lip)	5.0* / 2.5* (NS†)	2.6* / 1.6* (.03†)
Pg' (soft-tissue chin)	**7.9* / 4.0*** (.01)	5.6* / 1.8* (.01)
B point	5.1* / 2.7* (.04)	2.2 / 0.2 (NS)

Sign convention not printed; the text reads these as forward and down. † Recomputed from the printed means and SDs, the two P values for Li (and for UIA) appear transposed (derived, checks.py).

- The lips did not move back: they came forward, but the nose and chin came forward more. The nose came forward **2.6 mm (M) vs 1.3 mm (F)** more than the upper lip; the chin **2.9 vs 1.5 mm** more than the lower lip (Pn-h − Ls-h, Pg'-h − Li-h; derived). At about 15 y, the authors expect males to be only **1 year past peak velocity**, with more nose and chin growth to come (p 456).

👀 Key results 3: perception and preference (Tables VI–VII)

% change perceived (VAS)	Ext / Non (P)	Male / Female (P)
Orthodontists	35.8 / 33.9 (.63)	36.6 / 34.2 (.59)
Laypeople	38.2 / 37.5 (.80)	39.2 / 37.3 (.57)
Combined	36.5 / 36.3 (.95)	38.3 / 35.6 (.45)

- All perceived changes significant (P < .01): **small to moderate, 33.9–39.2 %**. Orthodontists and laypeople agreed (**r = 0.89**, P < .001).
- Perception did not track the ceph: only **1 of 18** correlations was significant, **N'-Pn-Pg' with orthodontists, r = 0.34, P = .03** (they noticed the profile straightening).
- **Preference** (post / long-term / neither, %): orthodontists **37.5 / 36.0 / 26.5** (ext), **38.3 / 28.9 / 32.8** (non), **42.3 / 29.0 / 28.7** (males); laypeople **32.0 / 38.6 / 29.5** (ext), **32.5 / 32.9 / 34.6** (non). About **30 / 30 / 30** overall: orthodontists leaned to the posttreatment photo, laypeople to the long-term one (except males); none significant. Variation was high (Table VII SDs 12.4–26.5): enough "to support any bias" (p 456).
- For **87.5 %** of patients preferences were clear, but not consistent across cases (one case ~90 % of orthodontists preferred long-term, another ~80 % posttreatment).

🧠 Main conclusions (paper's 5)

1. Treated to the **same incisor position and lip line**, extraction and nonextraction patients **stayed alike** 15 years later: the long-term change was not due to the type of treatment.
2. Between **15 and 30 years** the lips became **relatively retrusive** to the E- and S-lines; H-line did not change, so the authors attribute it mainly to **nose and chin growth**.
3. **Males changed more** than females: normal sexual dimorphism in soft-tissue aging, especially nose and chin.
4. Small-to-moderate change was perceived, equally by orthodontists and laypeople, with **no correlation** to the ceph change.
5. **No pattern of preference**; clear preferences for 87 % of profiles, but nothing showed what drove them.

⚠️ Limitations (our reading)

- **Matched at posttreatment**, with no pretreatment data: it tests whether extraction changes **aging**, not whether extraction **flattens the face**. Paquette 1992 matched **pretreatment** and found nonextraction faces about **2 mm fuller** (p 454), still present about 14.5 years out of retention (VERIFIED, abstract).
- **Small:** 20 per group; only differences of about **d 0.9** (≈1.5–1.9 mm in lip change) were detectable (derived). "No difference" ≠ equivalence: 95 % CI of the LS-E change difference **−1.22 to 1.42 mm** (derived).
- **No untreated controls**; white patients, one practice, finished "in the normal range"; Class I/II split, which premolars and retention not reported.
- Raters could see the age difference; photo standardisation and the ceph superimposition are not described; about 200 tests (derived count) without correction.
- **Reporting notes:** convexity wording (box above); the text gives ~30 / 30 / 30 where Table VII ranges 26.5–42.3 %; the abstract says raters judged among 40 profiles, each saw 20.

🔁 Other studies and Proffit

- **Agree:** Zierhut 2000 (no posttreatment difference; both flattened with nasal and mandibular growth), Luppanapornlarp 1993, Bravo Vallejo 2024 (similar attractiveness about 35 years after treatment), Formby 1994 (untreated, 18–42 y: male lips retrude). All VERIFIED, abstracts. Cited in the paper (p 456): Boley 1998, dental professionals could tell an extraction case only **50 %** of the time.
- **Treatment effect itself:** Konstantonis 2018 SR: extraction retracts the **lower lip 1.96 mm**, **upper 1.26 mm** more, nasolabial angle **+4.21°**; very low certainty (VERIFIED).
- **Proffit (VERIFIED, study notes):** Ch 2, the nose and chin grow more prominent after adolescence, the lips do not, so plan lip support at the end of treatment. Ch 7: comparable extraction and nonextraction cases differ by about 2 mm of lip fullness; the borderline decision rests on crowding, incisor position and the nose–lip–chin profile.

🦷 Clinical implications

- Decide extraction on the **target incisor and lip position**, not on fear of later "dishing in": finished alike, both groups aged alike.
- **On average both groups flattened** after treatment: about **1.5 mm** more relative upper-lip retrusion to the E-line over 15 years (Table III), **3.2 mm in males vs 0.9 mm in females** (Table IV); individual faces varied (SD up to 2.5 mm). Avoid finishing a teenage boy's lips already flat (our inference).
- Ceph numbers are not what people notice; a profile can age better or worse whatever the treatment.

✅ Number	Meaning
20 / 20 (6 M + 14 F each)	extraction / nonextraction
≈ 15 → 30 y	ages; interval ≈ 15 y (group means 14.1–15.0)
U1-SN, L1-NB, H-line	matching variables at posttreatment
−1.5 / −1.6 mm	upper lip to E-line change, ext / non (P = .82)
+4.0° / +2.9°	N'-Sn-Pg' change: profile **straighter**
−3.2 vs −0.9 mm	upper lip to E-line change, males vs females

✅ Number	Meaning
6.7 vs 3.1 · 7.9 vs 4.0 mm	nose (Pn-h) · chin (Pg'-h) forward, M vs F
33.9–39.2 % (≈36 % combined)	change perceived on the VAS (Table VI)
r = 0.89 · 1 of 18 (r = 0.34)	ortho–lay agreement · ceph–perception correlations
≈ 30 / 30 / 30 % · 87.5 %	preference split · patients with clear preferences